



*Division of Computing Science and Mathematics  
Faculty of Natural Sciences  
University of Stirling*

**Clinical NLP  
and Information Extraction from Electronic Health Records**

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**Dissertation submitted in partial fulfilment for the degree of  
Master of Science in Big Data**

**September 20xx**



## Abstract

Summary of the dissertation ***within half a page.***

Unnumbered chapter headings, such as above, are entered using the *Unnumbered 1* paragraph style. The *Unnumbered 1* style automatically starts a new page.

This template starts the page numbering at the foot of this page. While you are printing drafts, you might find it useful to add the printing date and time into the footer – to help you, and your supervisor, tell which version is most current.

It is suggested that the abstract be structured as follows:

- Problem: What you tackled, and why this needed a solution
- Methodology: How you went about solving the problem
- Results: Highlight your most important results.

## Attestation

I understand the nature of plagiarism, and I am aware of the University's academic integrity policy.

I certify that this dissertation reports original work by me during my MSc dissertation except for the following: *<examples given below: adjust according to your circumstances>*

- *The description of the methodology in Section <section number> was largely taken from [<citation>].*
- *The code discussed in Section <another section number> was created by <organisation> and used in accordance with the licence supplied.*
- *The code discussed in Section <yet another section number> was provided by my supervisor.*
- *The code discussed in Section <yet another section number> was developed by me during a vacation placement with the collaborating company. In addition, this used ideas I had already developed in my own time.*
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- *During development, the following tools were used: <list of AI tools>. In accordance with the AI policy for the MSc dissertation, all code generated by AI is marked as such, and the prompts are included with the submitted code.*

**Signature**      *(you must delete this, then sign and date this page)*

**Date**

## **Acknowledgements**

Acknowledge anyone that you wish to thank who has helped you in your work or supported you in any way: such as your supervisor, technical support staff, fellow students, external organisations or family. Acknowledge the source of any work that is not your own.

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The table of contents below is automatically generated from the paragraphs of style *Heading N* and *Unnumbered N*. To update this after revisions, right-click in the table and choose *Update Field* for the entire table; you may also be prompted about updates when printing.

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# 1 Introduction

For editorial consistency, it is important to use Word styles properly. Word has so-called quick styles. If the styles referred to below are not visible on the *Home* ribbon in the *Styles* category, choose *Apply Styles* from the down arrow at the bottom right of the *Styles* category. Styles can then be applied from the drop-down box. To make a style visible as a quick style, choose *Apply Styles*, then click *Styles* (the AA icon), then click on the drop-down list for a style, and then *Add to Quick Style Gallery*.

Chapter headings, as at the top of this page, are entered using the Heading 1 paragraph style. The Heading 1 style automatically moves to the start of a new page, and supplies the next chapter number. The new paragraph when you press Return/Enter after a heading automatically uses the *Body First* paragraph style. These "normal" paragraphs use 11 point Calibri font with single line spacing.

Subsequent paragraphs use the Body Text paragraph style. Enter most text using the Body Text paragraph style. The new paragraph when you press Return/Enter after a Body First paragraph automatically uses the Body Text paragraph style.

In general, use the default spacing that headings and paragraphs give you. Avoid using extra new-lines or spaces to format text. If you wish to emphasise something, usually use *italic font*.

## 1.1 Background and Context

Give the background to your project and context of what you have done. Section headings like this are entered using the *Heading 2* paragraph style – the *Heading 2* style automatically supplies the next section number, and the subsequent paragraphs are automatically Body First then Body Text.

## 1.2 Research Questions

List your research questions (typically two or three).

## 1.3 Scope and Objectives

Define the scope and objectives of your project, in order to answer your research questions.

## 1.4 Main Contribution

Summarise the main contribution that your work has made.

## 1.5 Overview of Dissertation

Briefly overview the contents of what follows in the dissertation.

## 2 State-of-the-Art

Summarise current knowledge and what others have done in the topic of your dissertation. Do not include extensive background material. Instead, focus on explaining the gap in existing work that your dissertation will address. You may use tables and figures if appropriate. Write for someone familiar with computing, but not necessarily expert in the particular topics of your project. Give references to other work by using *cross-references* to entries in the References section, like this [2].

### 3 Technical Chapters (change this to something appropriate)

**Note:** This part of the dissertation will normally be expanded to be *a series of chapters*.

The technical body of the dissertation consists of a number of chapters (just one here, but there will usually be more). These will typically consist of the following, *each of which may be one or more chapters*, as appropriate to your project – your supervisor will be able to advise on a suitable structure:

- Methodology / Design: Identify and briefly explain the methods that you have used in your practical work. This might include, for example, the datasets and algorithms that you have used, and why you chose these. Or, for a development type project, it may include identification of requirements, choice of languages/frameworks/libraries, and design of the front- and back-end to meet the requirements. In all cases, you should **justify the design decisions** that you have made, e.g. why did you use a particular dataset, a particular framework, or a particular user interface design? You must also **justify how you will evaluate your work**, e.g. which metrics will you use to measure performance and why, or how will you carry out user testing and why is this the most appropriate way?

Note that extensive screenshots of code should **not** be submitted, since you will submit your entire code base separately. You should instead describe your implementation at a high level, e.g. using diagrams or pseudocode (try to model this on how the papers that you have read and cited present their technical work). Occasional small snippets of code may be included if necessary to explain key parts of your work to the reader.

- Results: Present and analyse your key results, using figures and tables which you refer to and discuss in the text. This may include, for example, statistical analysis comparing machine learning algorithms, or an overview of software that you have built followed by a user evaluation study.

#### 3.1 First Section

Subdivide your text into sections.

##### 3.1.1 First Subsection

If necessary, also use subsections. Subsections are entered using the *Heading 3* paragraph style (all these heading styles are self-numbering).

##### 3.1.1.1 First Subsubsection

If you really need subsubsections, enter these using the *Heading 4* paragraph style – but this should be rare!

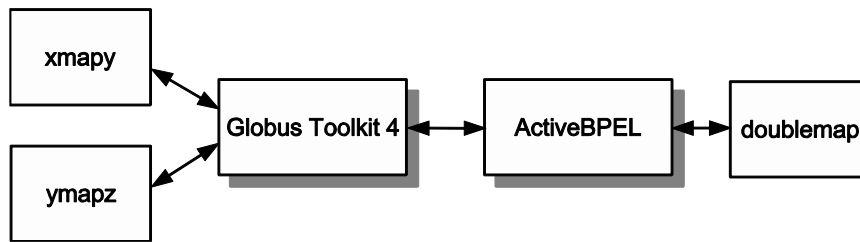
##### 3.1.2 Second Subsection

And, as required, more subsections.

#### 3.2 Second Section

As an example of a figure, consider Figure 1. Captions are entered using the *Figure* paragraph style. The figure below is placed in a *Body Centre* paragraph, which is set up in this document

to have an automatic *Figure* paragraph following it. *Figure* has automatic figure numbering, and it is possible to make *cross-references* to figures to refer to them from within the text. Move large figures to the top of the next page, *past any other text*, rather than having a big gap in the text.



**Figure 1. Highly Technical Diagram**

## 4 Evaluation and Conclusions

### 4.1 Summary

Summarise what you have achieved.

### 4.2 Evaluation of the Project

Stand back and evaluate what you have achieved and how well you have met the objectives and answered your research questions. Evaluate your achievements against your objectives in Section 1.3. Demonstrate that you have tackled the project in a professional manner.

(The previous paragraph demonstrates the use of automatic cross-references: The “3” is a *Cross-reference* to the text in a numbered item of the document, it is *not* literal text but a *field*. The number that appears here will change automatically if the number on the referred-to section is altered, for example if a chapter or section is added or deleted before it. Cross-references are entered using Word's **Insert** menu. Cross-references are set to update automatically when printed, but may not do so on-screen beforehand; you can update a field manually on-screen by right-clicking on it and selecting Update field from the pop-up menu.)

### 4.3 Comparison with Existing Work

Compare your results with those of work from your State-of-the-Art chapter. What have your results contributed to existing work?

### 4.4 Future Work

Explain any limitations in your results and how things might be improved. Discuss how your work might be developed further. This self-analysis is particularly important. You should give a critical evaluation of what went well, and what might be improved.

## References

Use the *Reference* paragraph style to enter and cross-reference document references. Books [1], standards [2], reports [3], journal articles [4], conference papers [5], and web pages [6] are conventionally presented in slightly different ways. You should follow the IEEE referencing standard. **Observe that all references should have a hyperlink to the work, as can be observed in the bibliography below.**

- [1] Greene, D. and Williams, P. C. *Linear Accelerators for Radiation Therapy*, Second Edition. IOP Publishing Ltd., Bristol and Philadelphia, 1997. Link: <https://doi.org/10.1201/9780429246562>
- [2] ISO. *Language Of Temporal Ordering Specification*, ISO 8807, International Organization for Standardization, Geneva, 1989. Link: <https://www.iso.org/standard/16258.html>
- [3] Jacobson, J. and Andersen, O., editors. *Software Controlled Medical Devices*. SP Report 1997:11, Swedish National Testing and Research Institute, Sweden, 1997. Link: <https://www.diva-portal.org/smash/get/diva2:962037/FULLTEXT01.pdf>
- [4] Turner, K. J. The Rules for Sailing Races on PDAs, *J. Navigation*, 23(5):114-240, May 2002. Link: <https://doi.org/10.1017/S0373463302001765>
- [5] Ji, H. and Turner, K. J. Specification and Verification of Synchronous Hardware using LOTOS. In Wu, J. Chanson, S. T. Gao, Q. editors, *Proc. Formal Methods for Protocol Engineering and Distributed Systems* (FORTE XII/PSTV XIX), pages 295-312, Kluwer Academic Publishers, London, UK, October 1999. Link: [https://doi.org/10.1007/978-0-387-35578-8\\_17](https://doi.org/10.1007/978-0-387-35578-8_17)
- [6] University of Stirling. Computing Science and Mathematics Research Home Page, <http://www.cs.stir.ac.uk/research>, April 2002.

## Appendix 1

You may have one or more appendices containing detail, bulky or reference material that is relevant though supplementary to the main text: perhaps additional specifications, tables or diagrams that would distract the reader if placed in the main part of the dissertation. Make sure that you place appropriate cross-references in the main text to direct the reader to the relevant appendices.

*Note that you should **not** include your program listings as an appendix or appendices. You should submit such text, and other project data/documents to the online repository following the instructions on Canvas.*